caugi: fast and flexible causal graph interface for R

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R package, causal inference

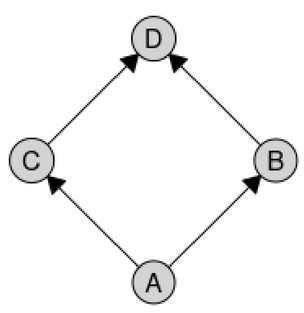
Motivation

- **Historically** users need to use adjacency matrices, edge lists, or general-purpose packages like `igraph`.
  - Writing a graph from an adjacency matrix is like writing a novel in binary: possible, but not fun.
- **`caugi` provides:** a *causality-first* graph package that emphasizes performance, flexibility, and readable syntax for causal graphs.

Basic Usage

Syntax like a picture in your head:

```
cg <- caugi(
  A %-->% B + C,
  B %-->% D,
  C %-->% D,#colbreak()
  class = "DAG"
)
plot(cg)
```



Querying and metrics

- **Relational queries**: parents(), ancestors(), neighbors(), etc.
- **Structural queries**: is_acyclic(), is_cpdag(), etc.
- **Graph manipulations**: add_edge(), remove_node(), etc.
- **Graph metrics**: shd(), aid(), etc.

How it works

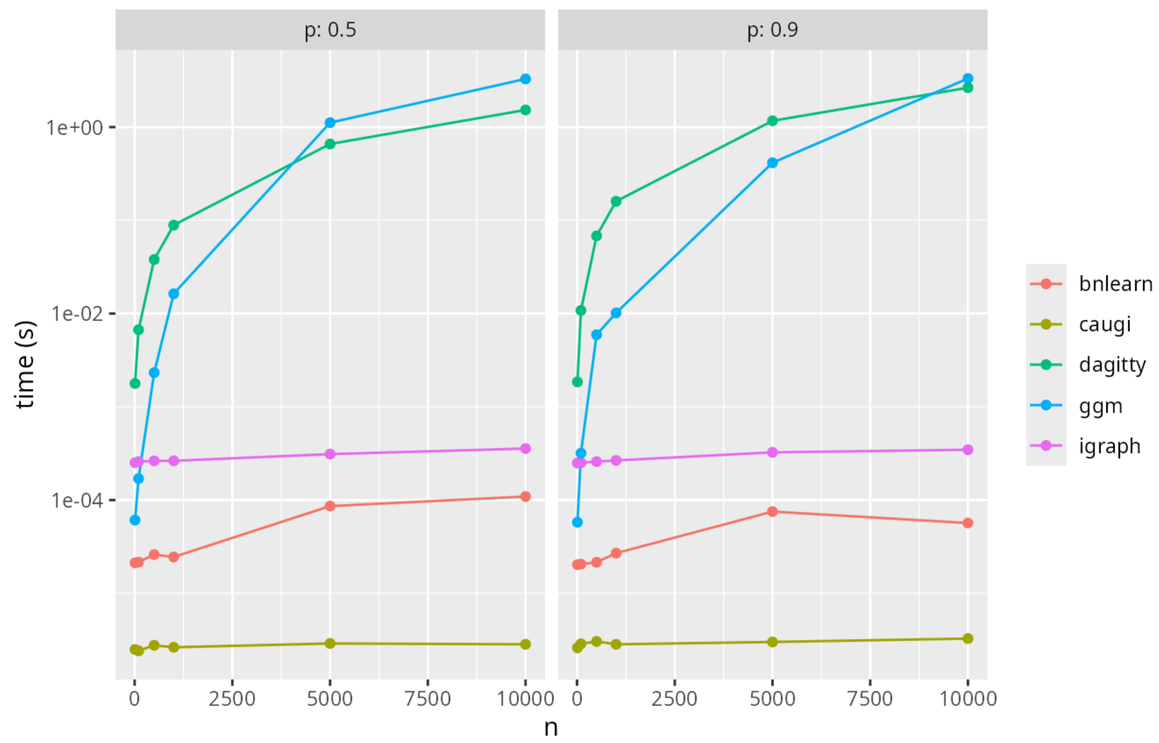
- **Backend**: Rust implementation for speed and safety.
- **Graph storage**: Compressed Sparse Row (CSR) format.
  - Fast queries; slower graph mutations compared to other formats.
- **Result**: Flexible and highly efficient causal graph querying in R.

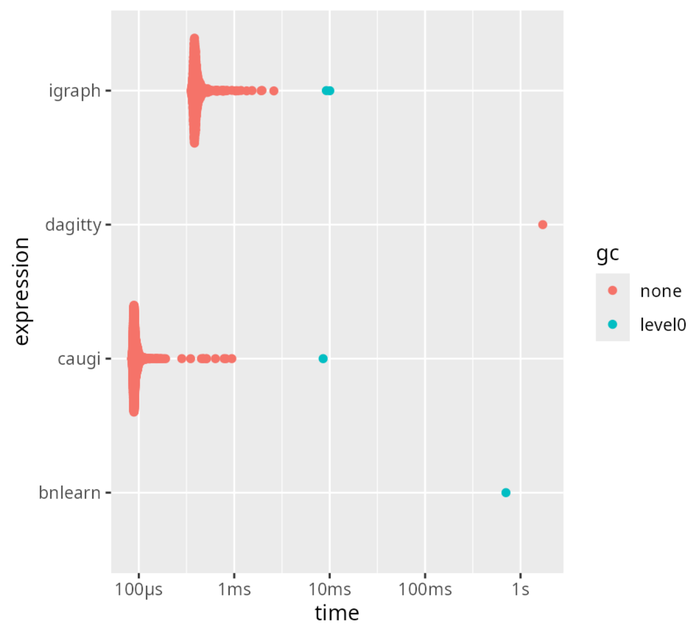
Why caugi?

- **Graph-class safe:** Think type safety, but for graphs.
  - Supports DAGs, PDAGs, ADMGs, UGs, and unknown graphs.
  - Trying to add an edge to a DAG that would create a cycle will throw an error.
- **Performance:** Fast querying of large graphs.
- **Intuitive syntax:** Express edges as simple R formulas, e.g., `A %<->% B + C`.

Benchmarks

We benchmarked `caugi` against `bnlearn`, `dagitty`, `ggm` and `igraph` for common graph queries on randomly generated DAGs of varying sizes.





Contact

- Pkgdown site: caugi.org
- GitHub: frederikfabriciusbjerre/caugi